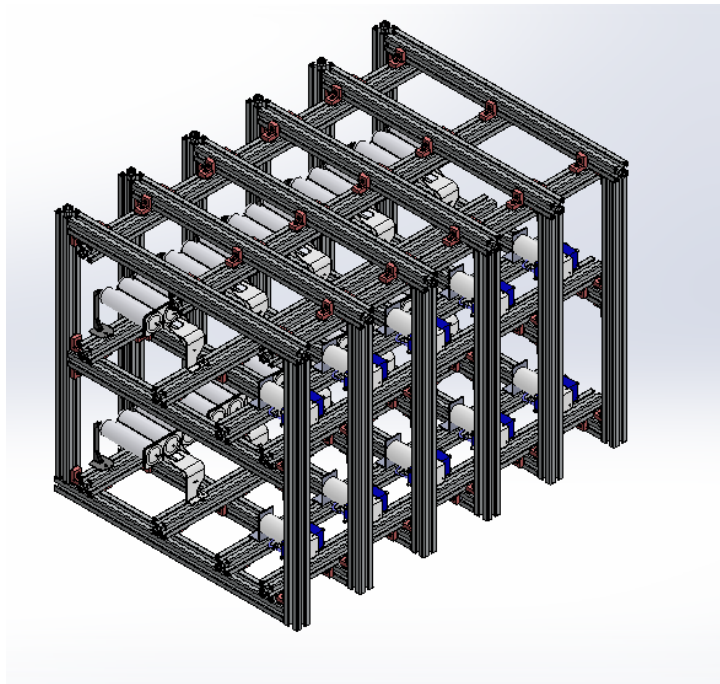


Automated Modular Pneumatic Reliability & Climatic Test System

Medical Device Component Qualification | Accelerated Life Testing (ALT) | FMEA



KEY SPECIFICATIONS & METRICS

Sample Size: 20 Units Under Test (2x 10-UUT modular racks in parallel chambers)

Fatigue Lifespan: 365,000 cyclic pressure pulses (Simulating 5 years / 1,825 treatments)

Test Duration: 45 Days continuous 24/7 operation

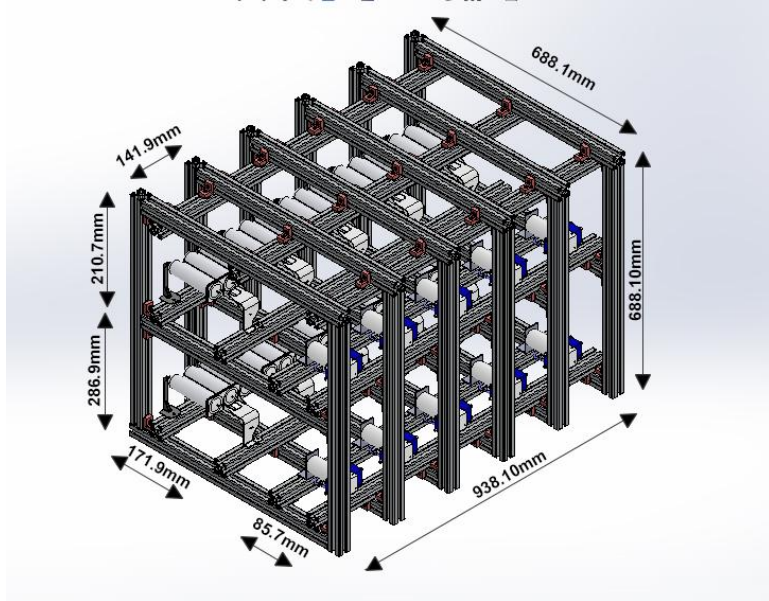
Test Conditions: 40°C controlled thermal environment

Acceptance Criteria: Leak decay < 3.4 mbar/min @ 2800 mbar, Pressurization < 40 s

Executive Summary:

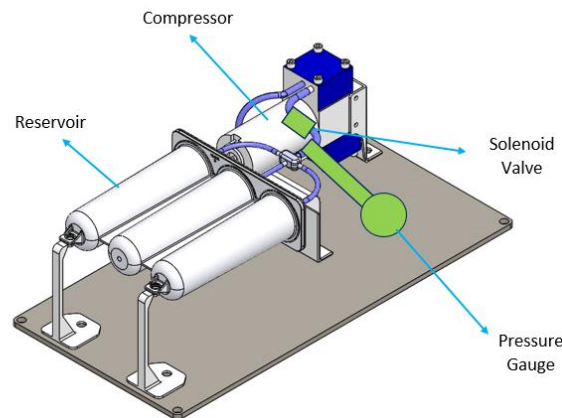
Designed, constructed, and commissioned a modular Accelerated Life Testing (ALT) system to qualify medical pneumatic sub-assemblies (air compressors, moulded reservoirs, and solenoids). To satisfy an n=20 statistical sample size within strict chamber volumetric limits, two identical 10-station racks were deployed in parallel across two 40°C thermal chambers for a continuous 45-day qualification run.

Mechanical Packaging, Modular Fixturing & Envelope Optimization



Chamber Envelope Optimization:

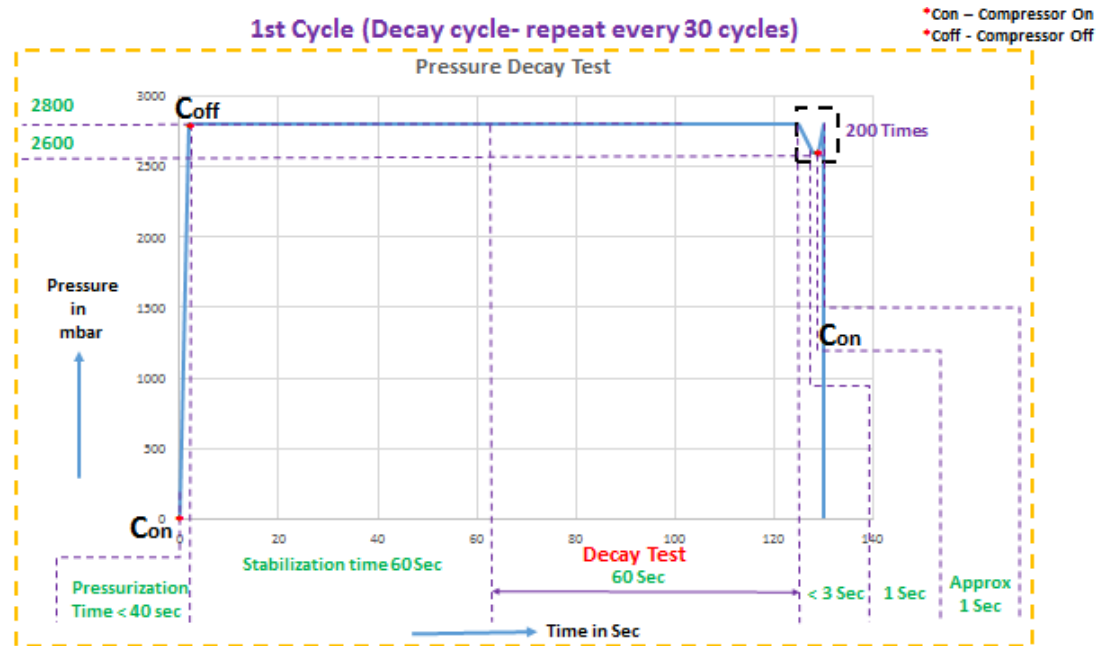
- Designed a rigid dual-tier structural frame using aluminum T-slot profiles.
- Sized precisely within internal chamber envelope limits (938.10 x 688.10 mm).
- Optimized tier spacing (286.9 mm lower tier / 210.7 mm upper tier) to maintain uniform thermal airflow around all active units.
- Calculated thermal expansion allowances for continuous 40°C operation.



Modular Quick-Swap Test Cells:

- Engineered dedicated single-station mounting plates for compressor, molded reservoir, solenoid valve, and sensors.
- Enabled independent single-unit removal and replacement in under 5 minutes without interrupting adjacent active tests.
- Integrated push-to-connect brass and polyurethane pneumatic fittings for quick servicing.

Pneumatic Closed-Loop Control, Arduino DAQ & Test Results



Pneumatic & Sensor Integration:

- Automated 24V power switching for cyclic compressor control.
- Generated high-frequency pressure pulses (2800 mbar <-> 2600 mbar for 200 repetitions per daily cycle).
- Integrated Arduino UNO and Data Logger Shield for continuous pressure decay and timestamped data logging.
- Central accumulator reservoir sized to buffer pressure spikes and maintain identical line pressure across all 10 stations.

Reliability Results & FMEA Verification:

- 5-Year Life Qualification: Successfully executed 365,000 pressure cycles over 45 continuous days.
- Leakage Threshold Verification: Captured automated 60-second pressure decay every 30 cycles to verify decay remained below 3.4 mbar/min.
- Failure Mode Mitigation: Isolated early seal micro-leakage modes at 40°C, providing corrective fitting design recommendations to R&D prior to high-volume production.